

# Risk Estimation of Preeclampsia

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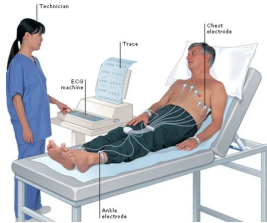
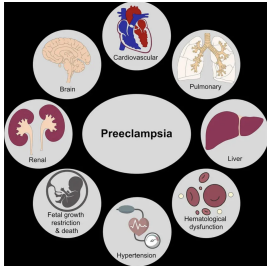
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# Preeclampsia Introduction

- ▶ Cannot be cured until birth
- ▶ Can progress to severe at a quick rate
- ▶ A leading cause of maternal and infant illness and death

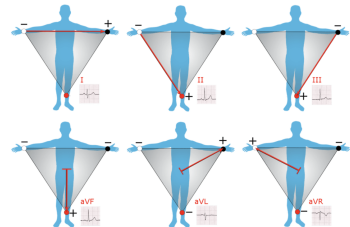
**Current Goal: Predict the risk of Preeclampsia using Decision Tree Modeling and Neural Network Modeling**



## Precordial Leads



## Limb Leads



## ECG Dataset Breakdown

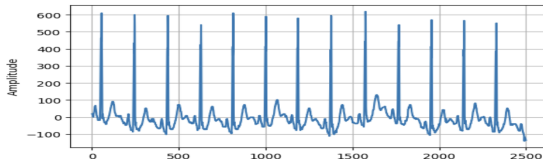
| Dataset    | Original Total<br>(P/N) | Dropped ECGs<br>(P/N) | Final Total<br>(P/N) |
|------------|-------------------------|-----------------------|----------------------|
| Unbalanced | 337 / 1854              | 8 / 24                | 329 / 1830           |

Each ECG sample includes **656 distinct MUSE metadata features**.

## Cross-Validation Strategy:

- ▶ **StratifiedGroupKFold** with 5 folds grouped by the Patient's MRN
- ▶ **Ensemble** with **RandomUnderSampler** inside the folds

## ECG Signal



## Unweighted-Voting Model:

- ▶ **RandomForest**
- ▶ **CatBoost**
- ▶ **XGBoost**
- ▶ **LightGBM**
- ▶ **LogisticRegression**
- ▶ **SVM**

## Model Performance

| Model              | AUROC            | AUPRC            | Specificity      | Sensitivity       | Balanced Accuracy | F1-Score         |
|--------------------|------------------|------------------|------------------|-------------------|-------------------|------------------|
| Unweighted Voting+ | $75.72 \pm 3.94$ | $41.58 \pm 8.03$ | $74.64 \pm 4.70$ | $62.35 \pm 10.93$ | $68.50 \pm 3.22$  | $40.85 \pm 2.24$ |
| Neural Network     | $70.9 \pm 4.9$   | $34.2 \pm 7.5$   | $73.0 \pm 9.2$   | $62.7 \pm 12.6$   | $0.00 \pm 0.00$   | $40.4 \pm 6.9$   |

## Data & Validation

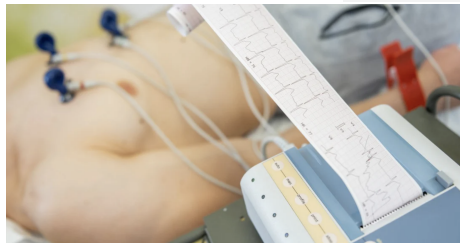
- ▶ Larger, multi-site datasets
- ▶ Temporal analysis
- ▶ Subgroup analysis

**End Goal: Make ECG diagnoses as easy and available as a thermometer!**



## Modeling & Deployment

- ▶ Hybrid ensemble
- ▶ Clinical decisions support
- ▶ Lightweight deployment (fewer leads)



# Thank you

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Questions?